

Course outline Winter 2025

Course Name: Programming Practices.

Course Code: CS – 2820

INSTRUCTOR

Instructor(s)	Dr. Hossam Mekawy
E-mail	Hossam.mekawy@uofcanada.edu.eg
Office Location	B211
Office Hours	(by appointment – please send an email a day before to check if the time slot is not already reserved) Sunday 9:30 AM-10:30 AM Thursday 10:00 AM-11:00 AM

TEACHING ASSISTANT

Instructor(s)	Eng. Omar Eltouney
E-mail	Omar.Aymen@uofcanada.edu.eg
Office Location	B318
Office Hours	Tuesday: 10-11 AM Wednesday: 1-2 PM

Class Time (subject to change)

Lecture	Lab
Thursday 8:30 pm - 09:45 am	Monday 1:40 PM – 2:55 PM
Thursday 1:40 pm - 02:55 pm	

Class LOCATION

Lecture	Lab
B102-A513	A513

COURSE DESCRIPTION

This course introduces the student to development in the Unix/Linux environment. Topics include development tools, shell programming, common utility programs, processes, file/directory management, IDEs, testing/debugging, version control, and an introduction to software engineering. Three lecture hours per week

Requisites:

The enrolled students must have passed successfully CS 152 (INTRODUCTION TO COMPUTER SCIENCE II) and CS 1920 (Programming Essentials)

RECOMMENDED TEXTBOOKS

Unix: The Textbook, 3/E by Syed Mansoor Sarwar and Robert Koretsky. Publisher: CRC press, Taylor \& Francis Group, 2017, ISBN-13: 978-1-4822-3358-2.

Recommended reading/additional resources:

Sed and Awk 101 Hacks: Enhance Your UNIX / Linux Life with Sed and Awk

(<https://www.thegeekstuff.com/sed-awk-101-hacks-ebook/>)

Any C programming reference book chosen by the student

Tools: The student is expected to create a Linux virtual machine or install Windows Subsystem for Linux (WSL) on his/her own laptops for hands-on activities off campus.

LEARNING OBJECTIVES

1. Explain Operating systems, their services, typical configurations, and classification based on the concurrency support for users and processes.
2. To briefly describe the UNIX operating system, its software architecture, history, different variations of the UNIX systems, the generic structure of UNIX commands and methods of connecting and logging on to a UNIX OS.
3. Explain the UNIX file concept, file types, attributes, file system organization, and access permission/privileges. Manage and maintain files and directories and perform file processing functions that include displaying file contents, compression, search, sort, RegEx, cut and paste, and encryption.
4. Describe what a UNIX shell is, identify shell types, some commonly used shells and their differences. Write and discuss basic Bourne (sh) and C (csh) shell scripts.
5. Explain the general utility of editing text files on a UNIX system and utilize the basic capabilities of vi, pico and emacs text editors. Perform basic programmable text processing using awk and sed.
6. Perform basic job control operations in UNIX.
7. Acquire basic knowledge of writing programs using C on Linux and compile, link and execute them. Develop basic understanding of different software development tools on UNIX such as make, gcc, and ar.

CLASS SCHEDULE AND GRADING SCHEME

<i>Week</i>	<i>Lecture</i>	<i>CLO</i>
1 (Feb 6 th)	Introduction to Operating Systems and UNIX - Overview of operating system - UNIX OS Software architecture - UNIX Kernel - User interfaces - Operating system types - UNIX OS development and history - Linux variation of UNIX - Differences between major UNIX families.	1
2 (Feb 13 th)	Files and Directory structure: - log on/off, - basic Files and Directory Structure commands, - UNIX File, and file system structure - Ordinary Files and Directory - Links (hard and soft) - Device files - Pipes and named pipes - Common structure of all UNIX flavors - Unix command structure - UNIX utility programs categories - ls, touch, echo, cat, more, less, cp, mkdir, mv, cd, pwd, rmdir, rm, man, whatis, whoami, whereis, which, w, ln, cal, date, write, talk, alias, unalias - redirection in Linux	2
3 (Feb 20 th)	links, file attributes and permissions, Process control - Access permissions. - Files and folders attributes.	3,4

	- Changing access permissions (chmod, umask) - User and group administration (List users, create new users and groups, change currently logged in users) - Change file ownership - SUID, SGID, and sticky bits - UNIX shell introduction	
4 (Feb 27 th)	Text file editing, basic file processing - Vi editor - Vi modes, starting options - Vi last line mode - Vi command mode - Search and replace (s command) - Metacharacters on search and replace. - Case sensitivity - Vi modifiers - Pico editor	5,3
4 (Feb 27 th)	Quiz 1	week 1-3 contents
5 (March 6 th)	Advanced file processing I - Regular expressions. - grep & egrep - find - sed	3
6 (March 13 th)	Midterm exam	Week 1-5 contents
7 (March 20 th)	Advanced file processing II - AWK and event driven scripting - AWK script execution - AWK relational expressions	3, 6

	- AWK range patterns and logical operators - AWK built in variables - Separator change - cutting and pasting - Sorting text files - File backup - encoding, decoding, - basic job control	
8 <i>(March 27th)</i> <i>(online / recorded)</i>	C programming review, - C language review - C program compilation, assembling and linking under Linux.	7
9 <i>(April 3rd)</i>	Quiz 2	Week 7,8 contents
9 <i>(April 3rd)</i>	Shell scripting - Available UNIX shells. - Changing login shells. - Executing shell scripts. - Variables in shell scripts - Reading from stdin - Passing arguments - Command substitution - Expression evaluation - Program flow commands (Test , Case, if, for, while, until)	4
10 <i>(April 10th)</i>	Midterm II	Week 7,8,9 contents
10	IDE, debugging and testing - Eclipse IDE	

(April 10 th)	- Debugging and testing in the IDE	
11 (April 17 th)	Development tools, version control, software engineering intro - Version control - software Engineering introduction - Make command and its options - o Makefile syntax - o Multiple targeting in makefiles. - o Multiple files c project using make.	7

GRADING SCHEME

Assessments	Grade	Date
Lecture participation	10%	All lectures
Assignment 1	10%	Week 3: Feb 20 th
Midterm 1	10%	Week 6: March 13 th
Midterm 2	10%	Week 10: April 10 th
Assignment 2	10%	Week 7: March 20 th
Assignment 3	10%	Week 11: April 17 th
Quiz1	5%	Week 4: Feb 27 th
Quiz2	5%	Week 9: April 3 rd
Final exam	30%	

HOMEWORK/ASSIGNMENTS

- You are expected to complete all weekly reading in preparation of class. Occasionally I may also hand out articles or cases not in the text that are required for you to read and prepare for class discussion. In addition, you will have individual/group assignments and term-based reports and assignments throughout the term. **ALL REQUIRED COURSE WORK MUST BE SUBMITTED TO PASS THE COURSE.**
- All reports required during this course shall be original and following academic integrity policy. The reports will be submitted on Turnitin tool, Turnitin check for similarities among other reports and the internet. If similarity beyond reasonable amount is detected penalties extending to filing an academic dishonesty case may be applied
- All students are expected to submit their work on time, late submissions without prior approved excuse will not be accepted.
- Students are required to use report templates if posted by the instructor.
- To support the student in acquiring the targeted learning outcome, this course has the following components:
 - Lectures.
 - Tutorials and Demos.
 - Assignments.
 - Exams and Quizzes.
- The tutorials and assignments in this course will offer thorough and diverse practices, demos, and exercises that resonate with the lectures' contents.
- Student's activities in the tutorials are graded.
- The student is expected to perform most of the course activities including exams and assignments on a Linux VM on their laptop or by accessing a shared Linux server on campus. Another option is to install Windows Subsystem for LINUX (WSL) on their Windows laptops.
- Bonus grades are typically attached to some activities.
- The 1st Midterm is typically conducted by the 6th week.
- The student is encouraged to utilize the office hours offered by the instructor and the TA to mitigate any difficulties encountered in his/her learning process.
- Bonus grades are typically attached to some activities and cannot exceed a total of 5%.

EXAMS

As per the below assessment dates table, generally, unless otherwise stated by the instructor, midterm exam/s (if applicable) will be held during the regularly scheduled lecture time.

ATTENDANCE AND PUNCTUALITY

Students in this course are required to attend all classes and be on time for all classes. There are no excuses for missing class. All absences are counted as time missed. This applies to all lectures, practical tutorials and/or labs. This also applies to mandatory office hours for probation & Gateway students. Students miss out on opportunities to acquire academic skills such as critical thinking, problem-solving, oral communication, teamwork, and decision-making when they fail to attend class. Being late for class disrupts the flow of the lecture and is disrespectful to the instructors and fellow students. Accordingly, students should note the following:

1. The instructors are not required to give substitute assignments or make up examinations for students who are absent.
2. It is the student's responsibility to make up the class content missed in consultation with his/her instructor(s). This does not mean that the instructor will provide a substitute class for the student.
3. Grades can be affected by missing a class as the instructor may include class participation marks and/or in-class assignments, or pop-up quizzes as part of the assessment plan for the course.
4. If a student is absent for 25% or more of the class time, tutorials, labs, and/or mandatory office hours, the instructor may:
 - a. Advise the student to drop the course if the absences occur before the drop date for course.
 - b. Give the student a grade of F for the course
 - c. Give the student an Incomplete grade for the course if the absences are caused by an approved documented illness or emergency. Students would then be given the opportunity to complete the course work as per the academic regulation.

ACADEMIC INTEGRITY

All students are expected to conduct themselves in all aspects of this course with the highest level of academic integrity. Accordingly, academic dishonesty will not be tolerated. Regulation 20 of the University Calendar defines academic dishonesty. Where there is reasonable evidence to support an allegation of academic dishonesty, Regulation 20 will be strictly enforced. All students are strongly advised to become familiar with this Regulation and UPEI Cairo Academic Integrity policy which can be found at: [!\[\]\(3d8c13c92b853674f749aac6fa869926_img.jpg\) UPEI Cairo academic integrity policy.pdf](#)

USE OF AI

The use of generative artificial intelligence (AI) and AI-assisted technologies is permitted only **upon receiving a written approval form the instructor**. Students should disclose in their **assignments** the use of AI and AI-assisted technologies in the writing process. Failure to do so will be considered as violation of the academic integrity policy.